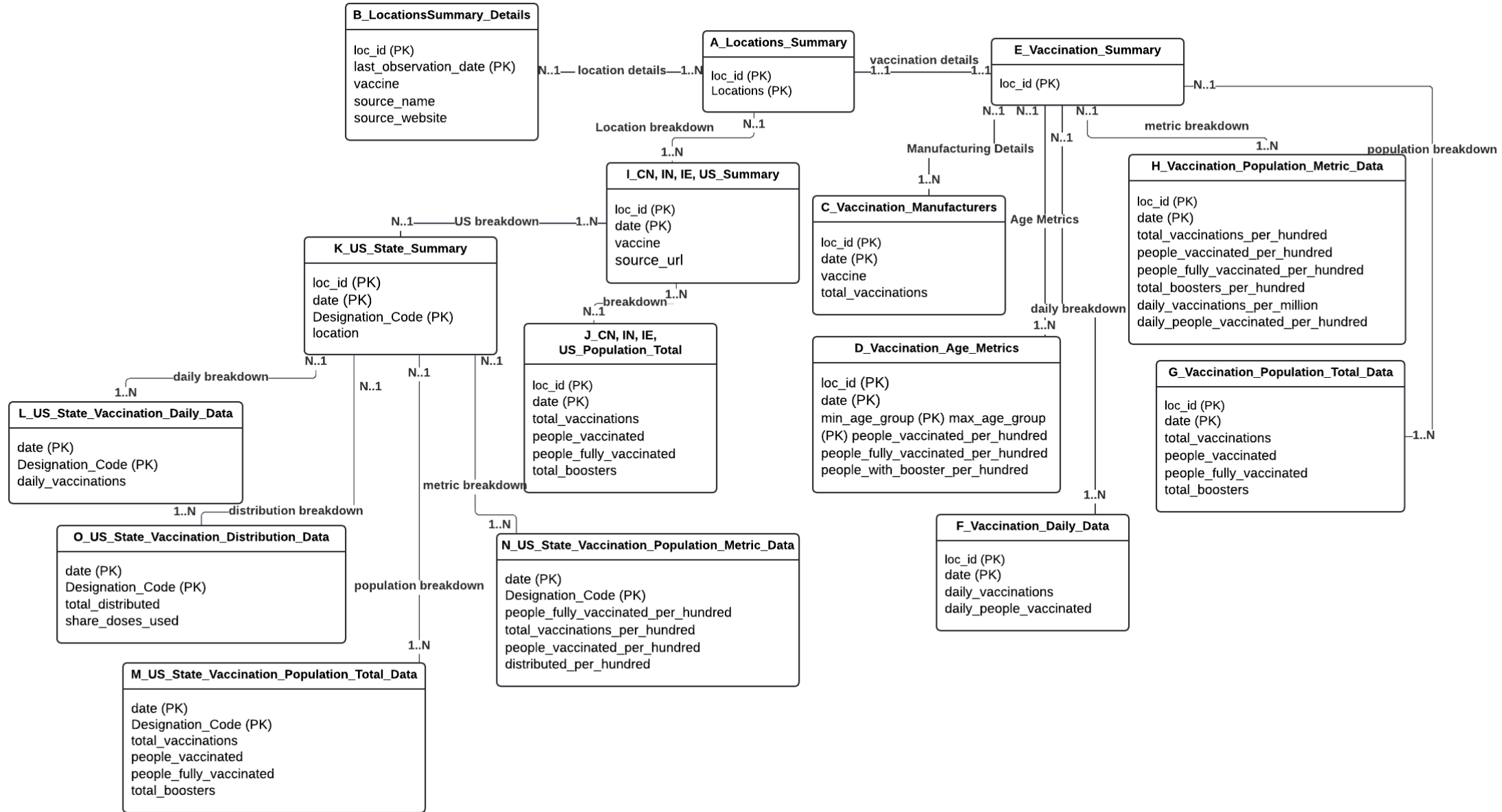


ER



Note: Data Cleaning Steps performed are at the bottom of the page

ER Schema

Step 1: Map Strong Entities

A_Locations_Summary (loc_id, location)

E_Vaccination_Summary (loc_id)

Step 2: Map Weak Entities

B_Locations_Summary_Details (loc_id*, last_observation_date, vaccine, source_name, source_url)

C_Vaccination_Manufacturers (loc_id*, date, vaccine, total_vaccinations)

D_Vaccination_Age_Metrics (loc_id*, date, min_age_group, max_age_group, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, people_with_booster_per_hundred)

F_Vaccination_Daily_Data (loc_id*, date, daily_vaccinations, daily_people_vaccinated)

G_Vaccination_Population_Total_Data (loc_id*, date, total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters)

H_Vaccination_Population_Metric_Data (loc_id*, date, total_vaccinations_per_hundred, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, total_boosters_per_hundred, daily_vaccinations_per_million, daily_people_vaccinated_per_hundred)

I_CN, IN, IE, US_Summary (loc_id*, date, vaccine, source_url)

J_CN, IN, IE, US_Population_Total (loc_id*, date, total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters)

K_US_State_Summary (loc_id*, date, Designation Code, location)

L_US_State_Vaccination_Daily_Data (date, Designation Code*, daily_vaccinations)

M_US_State_Vaccination_Population_Total_Data (date, Designation Code*, total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters)

N_US_State_Vaccination_Population_Metric_Data (date, Designation Code*, people_fully_vaccinated_per_hundred, total_vaccinations_per_hundred, people_vaccinated_per_hundred, distributed_per_hundred, daily_vaccinations_per_million, total_boosters_per_hundred)

O_US_State_Vaccination_Distribution_Data (date, Designation Code*, total_distributed, share_doses_used)

Step 3: Map 1:1 Relationships

A_Locations_Summary (loc_id, location)

E_Vaccination_Summary (loc_id)

Step 4: Map 1:N Relationships

B_Locations_Summary_Details (loc_id*, last_observation_date, vaccine, source_name, source_url)

C_Vaccination_Manufacturers (loc_id*, date, vaccine, total_vaccinations)

D_Vaccination_Age_Metrics (loc_id*, date, min_age_group, max_age_group, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, people_with_booster_per_hundred)

F_Vaccination_Daily_Data (loc_id*, date, daily_vaccinations, daily_people_vaccinated)

G_Vaccination_Population_Total_Data (loc_id*, date, total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters)

H_Vaccination_Population_Metric_Data (loc_id*, date, total_vaccinations_per_hundred, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, total_boosters_per_hundred, daily_vaccinations_per_million, daily_people_vaccinated_per_hundred)

I_CN, IN, IE, US_Summary (loc_id*, date, vaccine, source_url)

J_CN, IN, IE, US_Population_Total (loc_id*, date, total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters)

K_US_State_Summary (loc_id*, date, Designation_Code, location)

L_US_State_Vaccination_Daily_Data (date, Designation_Code*, daily_vaccinations)

M_US_State_Vaccination_Population_Total_Data (date, Designation_Code*, total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters)

N_US_State_Vaccination_Population_Metric_Data (date, Designation_Code*, people_fully_vaccinated_per_hundred, total_vaccinations_per_hundred, people_vaccinated_per_hundred, distributed_per_hundred, daily_vaccinations_per_million, total_boosters_per_hundred)

O_US_State_Vaccination_Distribution_Data (date, Designation_Code*, total_distributed, share_doses_used)

Step 5: Map M:N Relationships

Not present

Step 6: Multi-valued Attributes

Not present

Step 7: Map Higher-Degree Relationships

Not present

Final Schema – Using Functional Dependencies which served as my base

A_Locations_Summary loc_id -> location

E_Vaccination_Summary loc_id

B_Locations_Summary_Details (loc_id, last_observation_date) -> vaccine, source_name, source_url

C_Vaccination_Manufacturers (loc_id, date, vaccine) -> total_vaccinations

D_Vaccination_Age_Metrics (loc_id, date, min_age_group, max_age_group) -> people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, people_with_booster_per_hundred

F_Vaccination_Daily_Data (loc_id, date) -> daily_vaccinations, daily_people_vaccinated

G_Vaccination_Population_Total_Data (loc_id, date) -> total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters

H_Vaccination_Population_Metric_Data (loc_id, date) -> total_vaccinations_per_hundred, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, total_boosters_per_hundred, daily_vaccinations_per_million, daily_people_vaccinated_per_hundred

I_CN, IN, IE, US_Summary (loc_id, date, vaccine) -> source_url

J_CN, IN, IE, US_Population_Total (loc_id, date) -> total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters

K_US_State_Summary (loc_id, date, Designation_Code) -> location

L_US_State_Vaccination_Daily_Data (date, Designation_Code) -> daily_vaccinations

M_US_State_Vaccination_Population_Total_Data (date, Designation_Code) -> total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters

N_US_State_Vaccination_Population_Metric_Data (date, Designation_Code) -> people_fully_vaccinated_per_hundred, total_vaccinations_per_hundred, people_vaccinated_per_hundred, distributed_per_hundred, daily_vaccinations_per_million, total_boosters_per_hundred

O_US_State_Vaccination_Distribution_Data (date, Designation_Code) -> total_distributed, share_doses_used

Normal Forms

1st NF – tables have to be in atomic values. As Vaccine data was cleaned up, the tables met the requirements for 1st Normal Forms.

2nd NF - each non-key attribute is dependent on the primary key. Using Functional Dependencies to create tables helped ensure that all non-key attributes are dependent on all components of the composite Primary Keys. This was tested multiple times. Thus the tables meets requirements for 2nd Normal Forms

3rd NF - each attribute depends directly on the primary key or composite primary key. Tables were broken down according to their purpose and usefulness by using Decomposition method. The tables meets the requirements for 3rd Normal Forms.

Boyce-Codd Normal Form (Lei's Notes) - LHS is a candidate key. BCNF ensures tables are free from data redundancy. All functional dependencies have candidate keys on the LHS (left-hand side). Redundant were removed from each table and care was taken to ensure it wasn't excessively repeated/used without purpose. Thus is it meets the requirements of BCNF.

Data Cleaning Steps Performed

Task Performed	Reasoning
Creation of functional dependencies	Ensure tables are in their 3rd normal form

Tables are named alphabetically	For ease of reference and sorting
Counta Unique Excel function (removed from the database)	To check and ensure that foreign key relation matches the distinct count in tables
In all the income was removed but all other non-country regions were kept	Based on all available online data (I google searched this source), there isn't enough information to explain why non-country items were included separately in the tables. It could be due to variations in the data source or could be due to the need to analyse this differently. Based on my data knowledge and experience, I couldn't understand how can I merge or remove these items without properly understanding their source or use. Thus, I kept them for their possible usability if required. My apologies if I didn't dig enough into it.
A few items were renamed & US States (which includes regions related to US) were given a Code	For ease of referencing
The column 'daily_vaccinations_raw' was deleted	Based on the information available, it is recommended to use the daily_vaccinations column for analysis. Thus to eliminate possibility of duplication
Tables were broken down	Considering the amount of columns present, I felt it would make sense to break it down into tables for usability. It also minimises the size of data stored in the table, so the transfer of information can be made faster & easier
Blank & NULL columns were renamed to '0'	Given the purpose of this database (for this assignment) & that SQL queries were supposed to be generated with Visualisations, having blank or NULL values would increase the complexity of data processing given the database management source we're using.
Foreign keys were not included in DDL syntax but was manually updated	Due to the limitations of SQL Lite software, only primary keys were included in DDL while all other integrity constraints were manually updated

R Studio and Excel functions were both used	For data cleaning and to ensure consistency. This minimises human error and efforts (rechecking functions performed on the systems) were taken to ensure that systems errors were also minimised with the use of functions
Text to Column function in Excel & PIVOT Long in R Studio	The Vaccines were broken down into atomic values. Further minor enhancements were made to ensure the cleanliness of the data
Manufacturer's age is broken down into max and min age multiple tidying functions in Excel	I considered binning the age but I couldn't find the logic to it. I noted that one of the database uploaded by the same person used max and min age. Also based on Google, the maximum age of a lived person was 120 so I used as max age for figures where it had '+' sign like '100+'
North America(NAM) named to NAC and South America(SAM) named to SAC.New York State was given as designation code	Namibia also used NAM as a ID thus I changed both the North and South America to make them similar. Added an ID to New York State as it was absent